## After completion of preparation materials, students should be able to:

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**1. Subclasses**

Identify subclasses of the individual adrenergic blockers based on their receptor selectivity.

**2. Pharmacokinetic properties → Patient care**

List to ADME properties of starred drugs and their relevance to patient care.

**3. Epinephrine effect → Clinical rationale**

Explain the effects and clinical relevance of epinephrine reversal and compare to those of phenylephrine.

**4. Mechanisms of action → Cardiovascular effects**

Characterize the cardiovascular effects and reflex responses with administration of each of the following as monotherapy by giving examples of each: a nonselective alpha blocker, a selective alpha-1 blocker, a selective alpha-2 agonist, a nonselective beta-blocker, a selective beta-1 blocker, and a beta blocker with combined $\beta$ and $\alpha$ blocking actions.

**5. Mechanisms of action → Effects → Therapeutic uses**

Correlate the mechanisms of action and pharmacological effects of the adrenergic receptor antagonists to the agents' specific therapeutic uses.

**6. Mechanisms → Adverse effects → Cautious use**

Relate the mechanisms of adverse effects and potential drug interactions of the individual agents to their cautions and contraindications for use.

# What you need to know and understand about adrenergic blockers



- The selective alpha-1 blockers prazosin, terazosin, and doxazosin (quinazolines) are adjunctive (add-on) agents in the management of chronic primary hypertension or urinary obstruction caused by benign prostatic hyperplasia (BPH) and other conditions.
- The major adverse effect of these drugs is marked orthostatic hypotension with the first dose (first-dose effect). Relaxation of iris dilator muscle due to $\alpha_1$ antagonism can cause intraoperative floppy iris syndrome. The drugs can also cause nasal stuffiness due to alpha blockade in the vasculature of the nasal mucosa, and impaired ejaculation.
- Tamsulosin and silodosin are selective $\alpha_{1A}$ receptor inhibitors used for the management of urinary symptoms of BPH. Alfuzosin is a qinazoline $\alpha_{1A}$ receptor inhibitor for symptomatic treatment of BPH. These drugs are not used for hypertension.
- Yohimbine is an alkaloid derived from a botanical source. It is an $\alpha_2$ receptor competitive antagonist with effects opposite those of the $\alpha_2$ receptor agonist clonidine. It is available in herbal products for erectile dysfunction (never clearly demonstrated in clinical trials).

# What you need to know and understand about adrenergic blockers

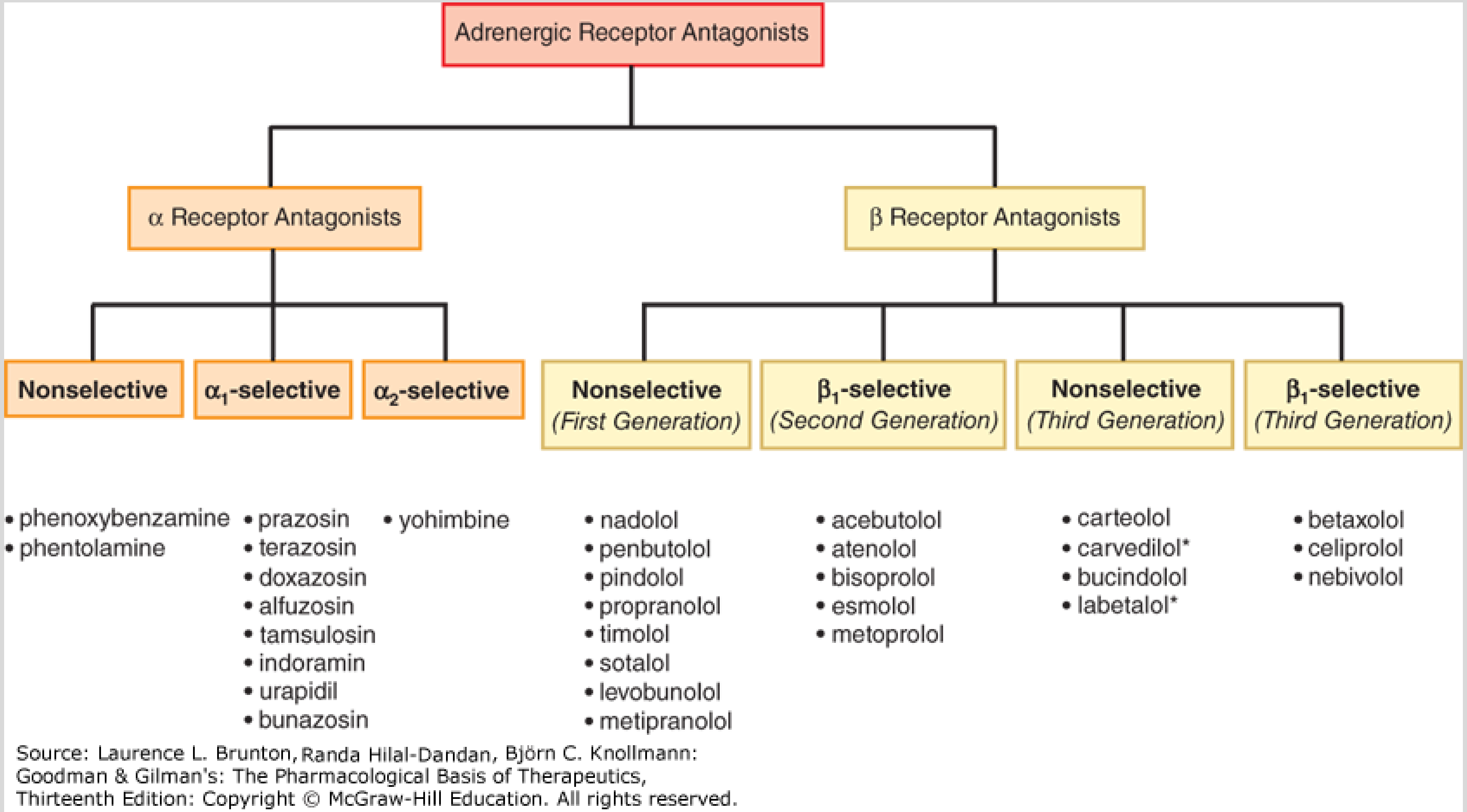


- Beta blockers antagonize the effects of catecholamines at β-adrenergic receptors. Evidence suggests that some agents are inverse agonists.
- The beta blockers differ in their relative affinities for βARs. The nonselective beta blockers inhibit both $\beta_1$ and $\beta_2$ receptors, whereas the selective agents inhibit $\beta_1$ARs at usual therapeutic doses. None of the clinically available $\beta_1$-selective agents is completely selective. Selectivity tends to diminish at higher drug concentrations.
- Some actions of beta blockers may be due to additional effects, such as partial agonist action, $\alpha_1$AR blockade, and membrane stabilizing effect.
- The beta blockers are well absorbed from the GI tract and undergo extensive hepatic metabolism. First-pass effect varies among the individual agents. They are rapidly and widely distributed to the tissues. Lipophilic beta blockers, propranolol, metoprolol, and nebivolol, distribute to the CNS and may cause CNS effects, such as fatigue, depression, and insomnia. CNS effects may contribute to blood pressure lowering effects.

# What you need to know and understand about adrenergic blockers



- Beta blockers have prominent effects on the heart, decreasing heart rate (chronotropy) and force of contraction (inotropy), and reducing the rate of conduction through the AV node. They inhibit $\beta_1$ receptors on specialized cells in the kidney, which suppresses renin release. (Renin increases blood pressure by increasing angiotensin II and aldosterone – the renin, angiotensin, aldosterone system, RAAS.)
- Beta blockers have a variety of clinical applications, including the treatment of hypertension, angina pectoris, myocardial infarction, cardiac arrhythmias, congestive heart failure, glaucoma, and the prevention of migraine headache.
- Mechanism-related adverse effects include bradycardia, bronchoconstriction, potentiation or masking of hypoglycemia, and lipid abnormalities.
- Abrupt withdrawal of beta blocker therapy should be avoided because it can cause severe rebound hypertension.




Adrenergic antagonists inhibit the interaction of endogenous NE and epinephrine, and also sympathomimetic drugs, with adrenergic receptors at effector organs.

Goodman & Gilman's The Pharmacological Basis of Therapeutics, 13e, 2018; Figure 12-3

# Antagonism of $\alpha$ARs:

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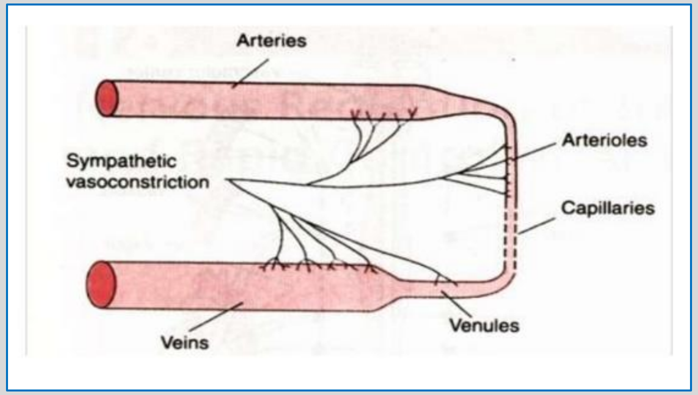
**Selective and nonselective alpha blockers block $\alpha_1$** receptor-mediated contraction of arterial, venous, and visceral smooth muscle.




**Nonselective alpha blockers also block $\alpha_2$** receptor-mediated presynaptic and postsynaptic actions.

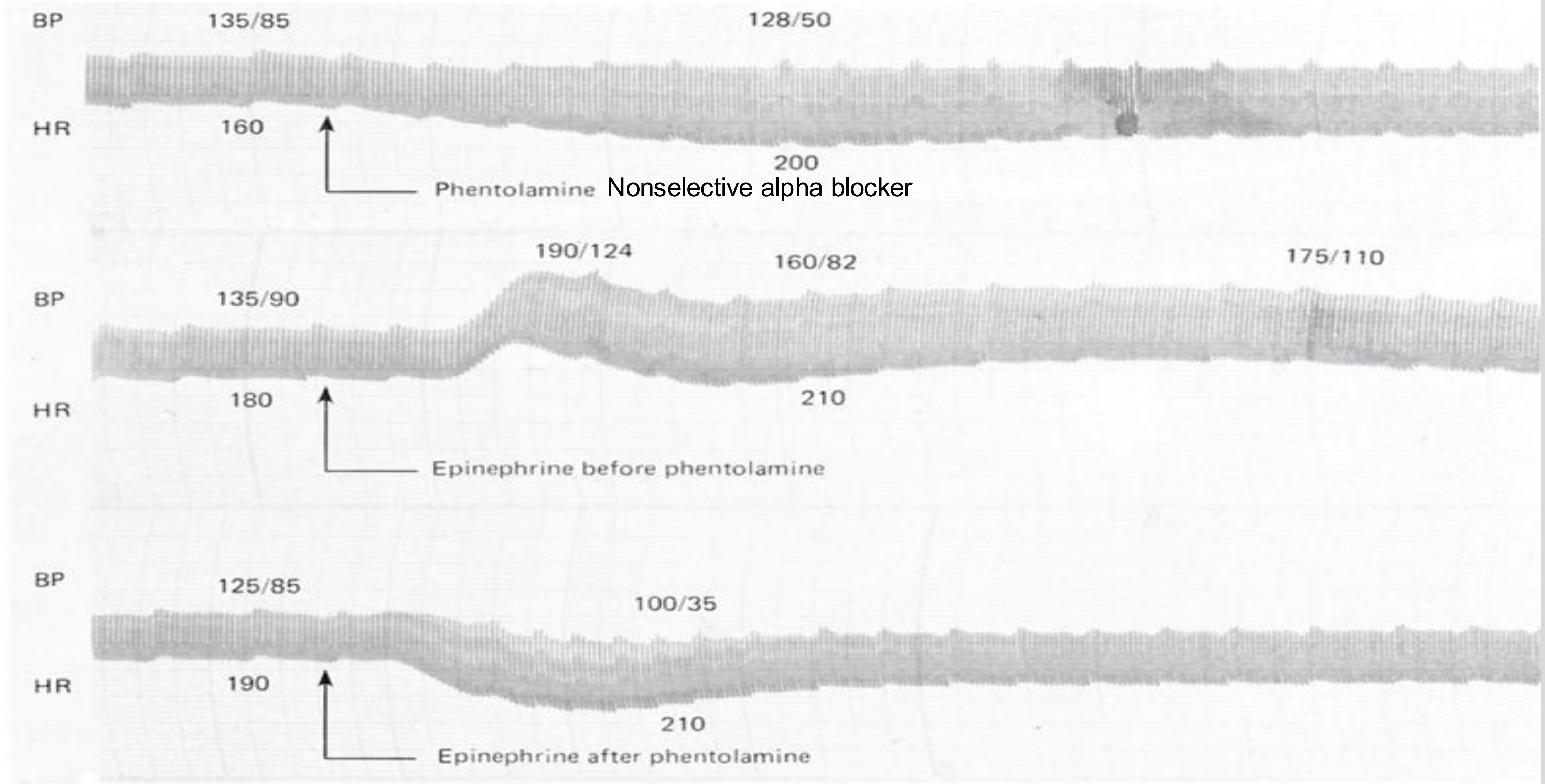
$\alpha_2$ actions are:

- suppression of sympathetic output
- increased vagal tone
- platelet aggregation
- inhibition of the release of NE and ACh from nerve endings
- regulation of metabolic effects (suppression of insulin secretion and inhibition of lipolysis)
- **contraction** of some arteries and veins by stimulation of post-synaptic $\alpha_2$ARs located on vascular smooth muscle.

All of these $\alpha$2-mediated actions are BLOCKED.

(yes, vasoconstriction but it is less pronounced effect than $\alpha_1$AR- mediated contraction)

# What is the phenomenon described by these tracings?




Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 12th edition*: www.accessmedicine.com

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Administration of an α blocker prior to admin of epinephrine converts EPI's pressor response into a depressor response. Inhibiting $\alpha$AR agonist-induced vasoconstriction allows EPI's unopposed $\beta$2 vasodilation $\rightarrow$ a reduction in blood pressure.

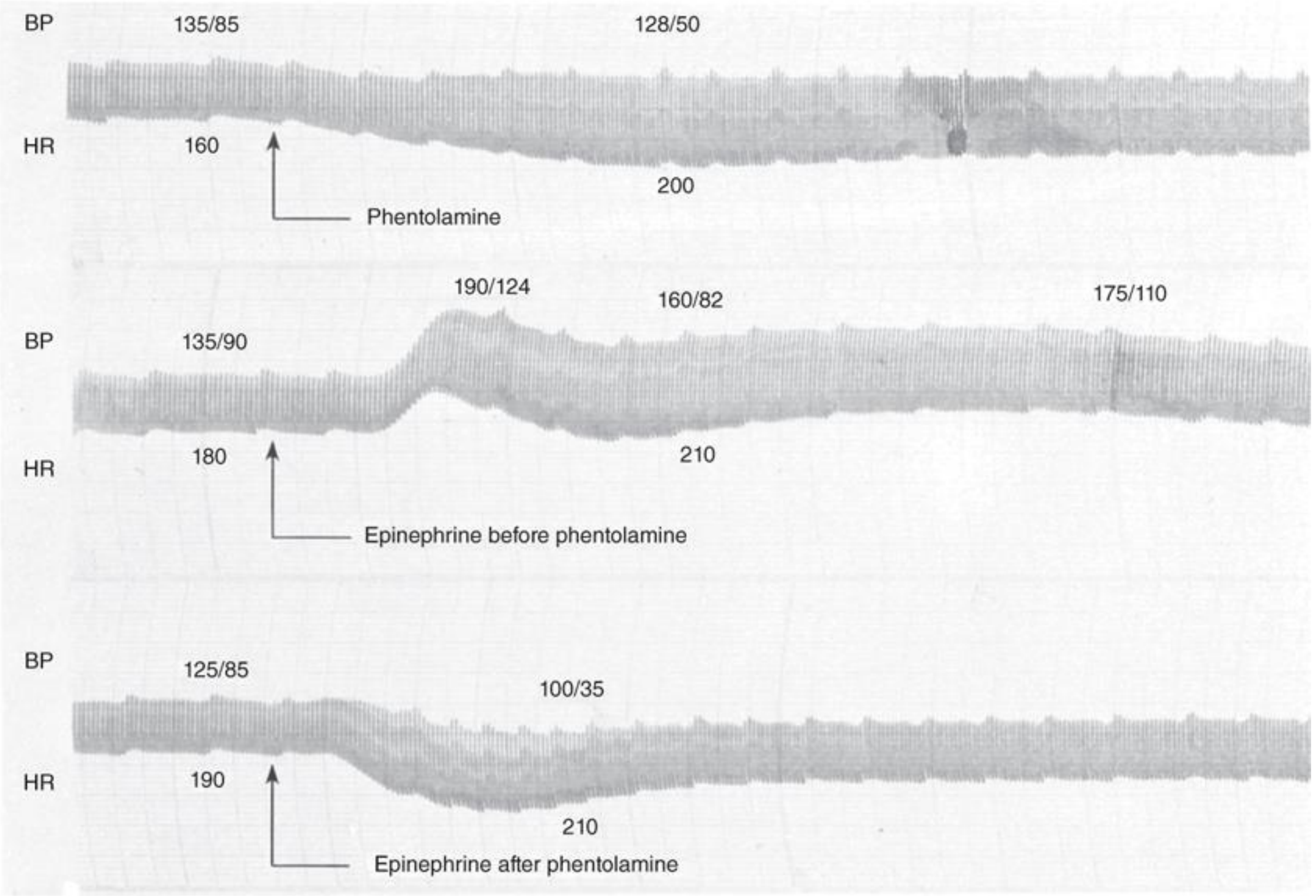
## Epinephrine Reversal

Top: Effects of phentolamine, an α-receptor-blocking drug, on blood pressure in an anesthetized dog.

Epinephrine reversal is demonstrated by tracings showing the response to epinephrine before (middle) and after (bottom) phentolamine.

(Intravenous administration)

BP, blood pressure; HR, heart rate.




Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Citation: Chapter 10 Adrenoceptor Antagonist Drugs, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595199 Accessed: December 29, 2020

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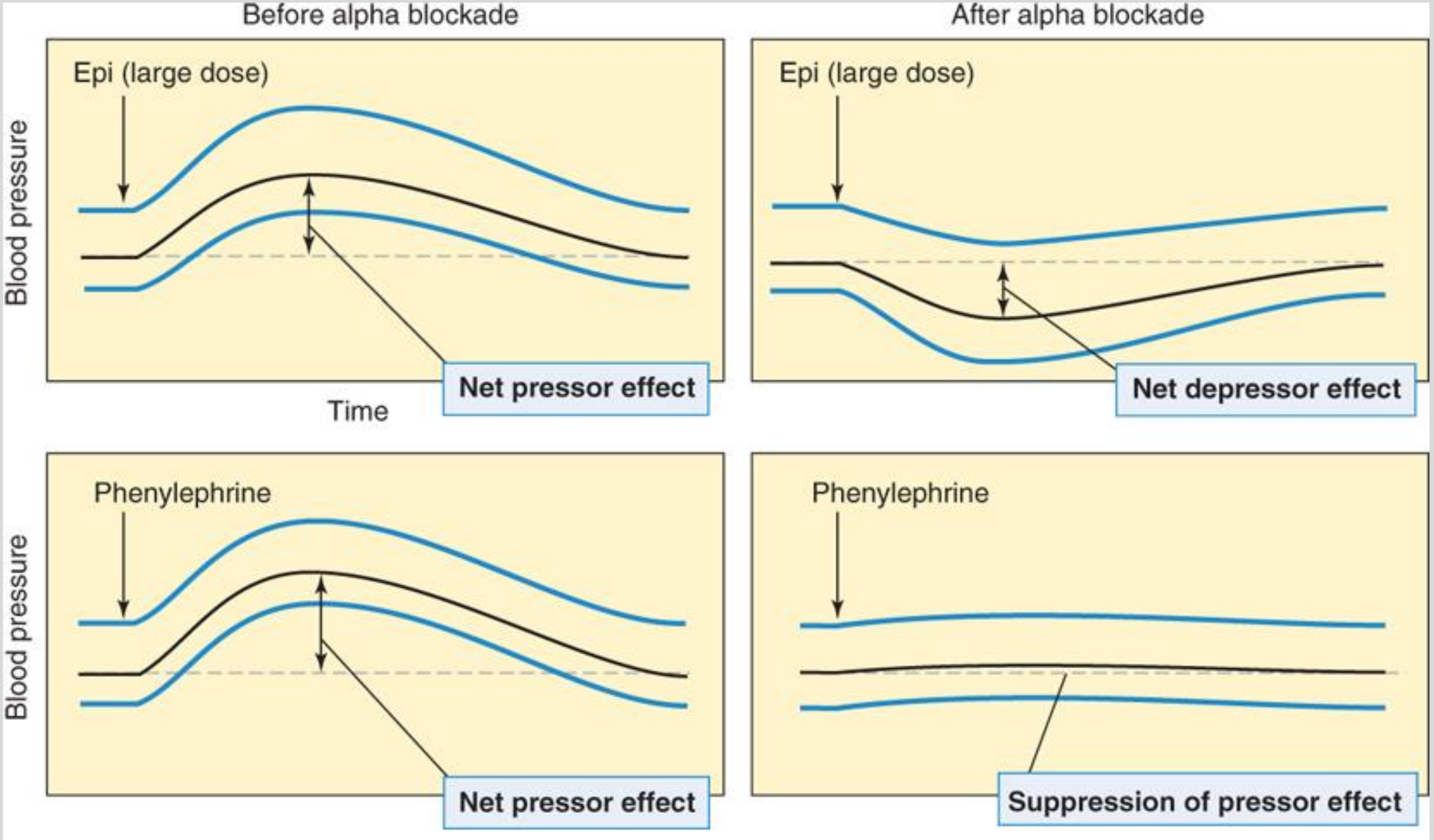
# Compare the responses of EPI and phenylephrine to alpha blockade.




Source: M. Kruidering-Hall, B. G. Katzung, R. L. Tuan, T. W. Vanderah: Katzung's Pharmacology Examination & Board Review, 14th Edition
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The effects of an α blocker, for example, **phentolamine**, on the blood pressure responses to **epinephrine** (epi) and **phenylephrine**.

- Top: The epinephrine response exhibits reversal of the mean blood pressure change from a net increase (the α response) to a net decrease (the $\beta_2$ response).
- Bottom: The response to phenylephrine is suppressed but not reversed, because phenylephrine lacks β action.

Citation: Chapter 10 Adrenoceptor Blockers, Marieke Kruidering-Hall, Bertram G. Katzung, Rupa Lalchandani Tuan, Todd W. Vanderah. *Katzung's Pharmacology Examination & Board Review, 14th Edition;* 2024. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285590454 Accessed: November 10, 2025
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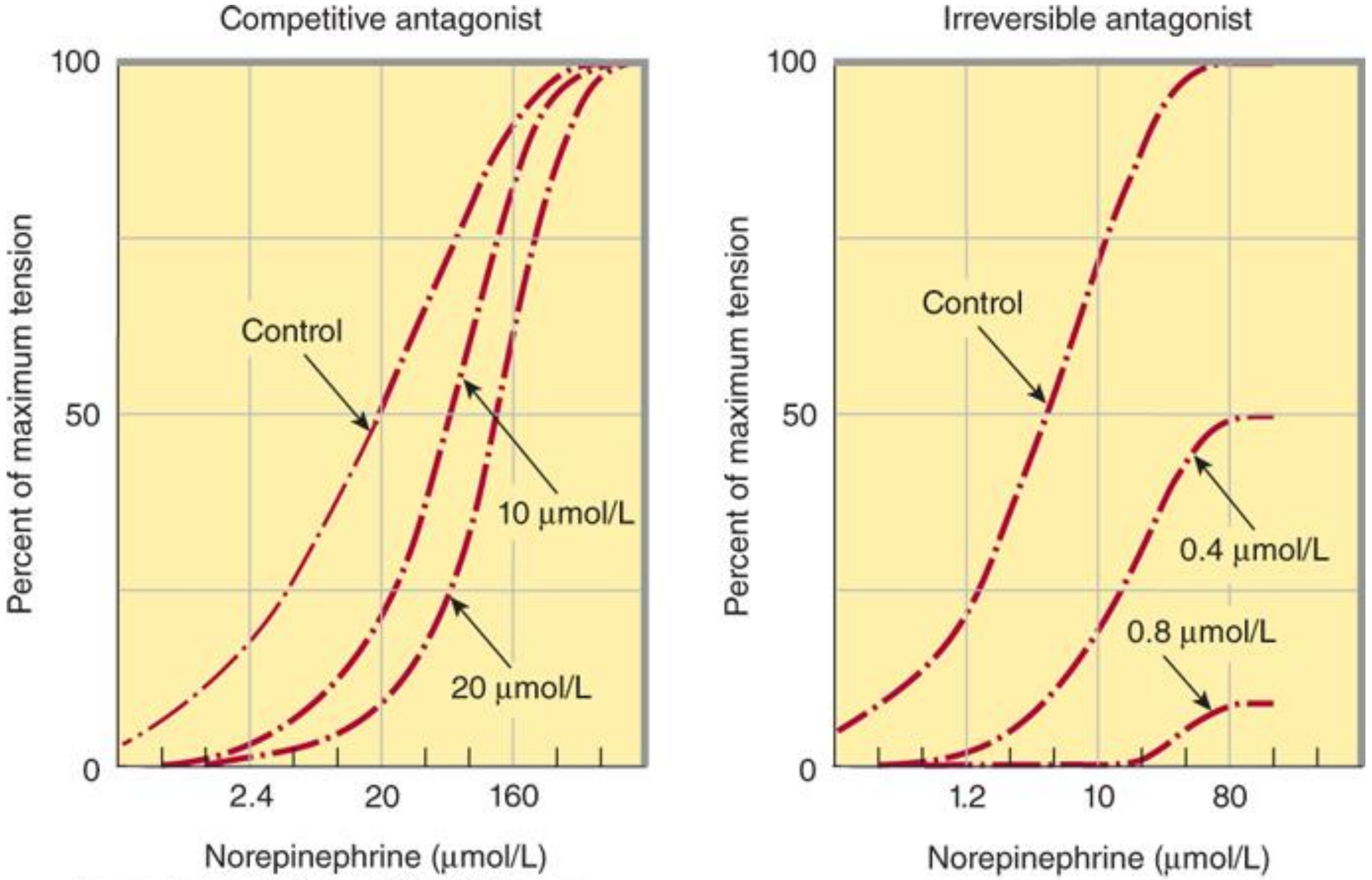
# Dose-response curves of norepinephrine in the presence of two different α-adrenoceptor–blocking drugs.

Left:

Tolazoline, a reversible blocker, shifted the curve to the right without decreasing the maximum response when present at concentrations of 10 and 20 μmol/L.

**Phentolamine** is a competitive, potent, nonselective α-adrenergic antagonist




Source: Bertram G. Katzung, Todd W. Vanderah: Basic & Clinical Pharmacology, Fifteenth Edition
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The tension produced in isolated strips of cat spleen, a tissue rich in α receptors, was measured in response to graded doses of norepinephrine.

(Adapted with permission from Bickerton RK: The response of isolated strips of cat spleen to sympathomimetic drugs and their antagonists, J Pharmacol Exp Ther 1963 Oct;142:99-110.)

Right:

Dibenamine, an analog of phenoxybenzamine and irreversible in its action, reduced the maximum response attainable at both concentrations tested.

**Phenoxybenzamine** is an irreversible αAR antagonist (some selectivity for $\alpha_1$ARs)

Citation: Chapter 10 Adrenoceptor Antagonist Drugs, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595199 Accessed: December 29, 2020
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# Clinical Applications of Alpha Blockers



**Antagonism of EPI and NE at $\alpha$ARs for:**

**Phenoxybenzamine**

- Presurgical management of hypertension and sweating associated with pheochromocytoma
- Treatment of urinary symptoms associated with benign prostatic hypertrophy

**Phentolamine**

- Intraoperative hemodynamic instability: Rapid, short-term control of acute hypertension in patients undergoing surgery for pheochromocytoma
- Extravasation of a pressor agent (infiltration)
- Reversal of oral soft tissue (dental) local anesthetic containing epinephrine for local vasoconstriction (infiltration)

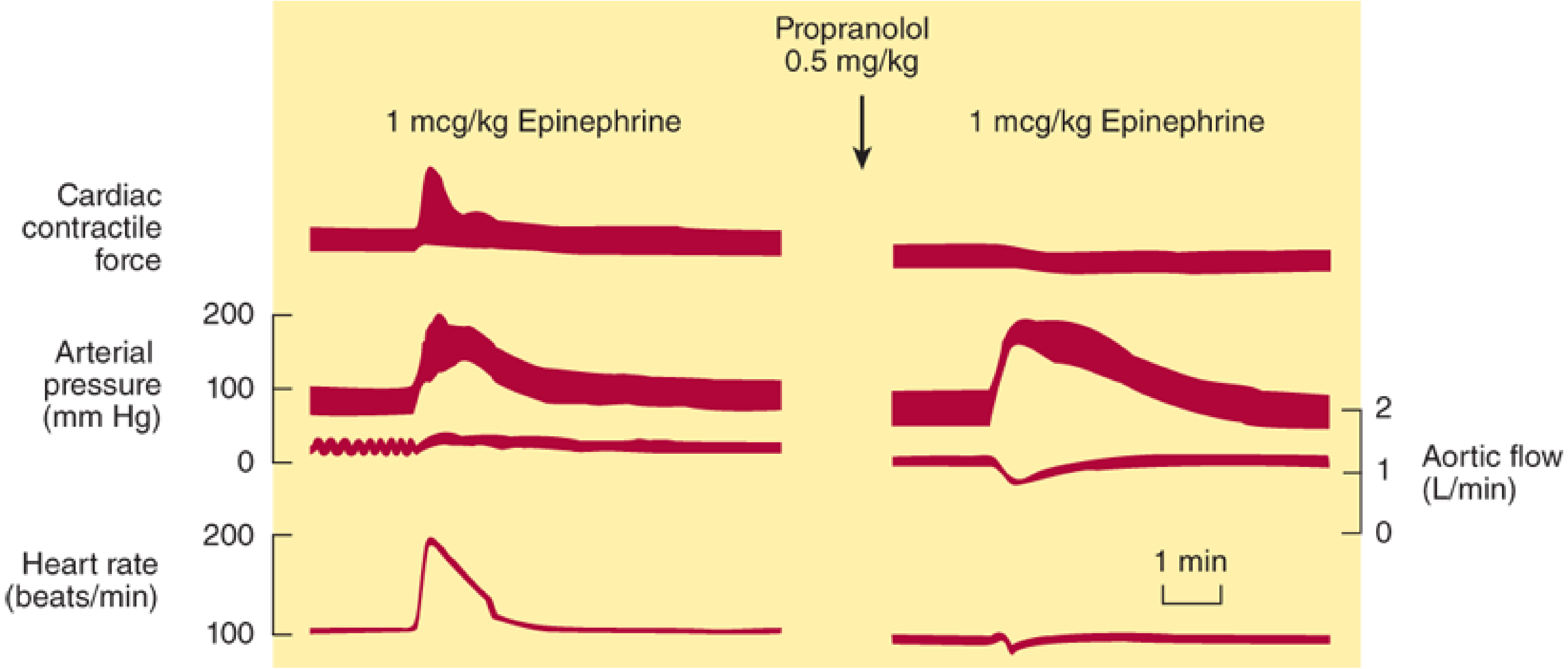
# Selective αAR Antagonists

| $\alpha_1$-Selective | | $\alpha_2$-Selective |
|---|---|---|
| Quinazolines for the treatment of HTN | Drugs for treatment of urinary symptoms of BPH | |
| ✳Prazosin<br>• Terazosin<br>• Doxazosin<br><br>These alpha blockers are also used for patients with hypertension and concomitant urinary symptoms due to BPH. | ✳Tamsulosin ($\alpha_{1A}$ blocker)<br>• Silodosin<br>• Alfuzosin<br><br>These drugs are NOT used in the treatment of hypertension. | ✳Yohimbine (botanical product) |

## BB Cardiovascular Effects: In the presence of a βAR-blocking agent, EPI no longer augments the force of contraction nor increases cardiac rate.

## Blood pressure is still elevated by EPI because vasoconstriction is not blocked.

The effect in an anesthetized dog of the injection of epinephrine before and after propranolol. In the presence of a β-receptor–blocking agent, epinephrine no longer augments the force of contraction (measured by a strain gauge attached to the ventricular wall) nor increases cardiac rate. Blood pressure is still elevated by epinephrine because vasoconstriction is not blocked.

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(Reproduced with permission from Shanks RG: The pharmacology of beta sympathetic blockade, Am J Cardiol 1966 Sep;18(3):308-316.)

Citation: Chapter 10 Adrenoceptor Antagonist Drugs, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250595199 Accessed: December 29, 2020

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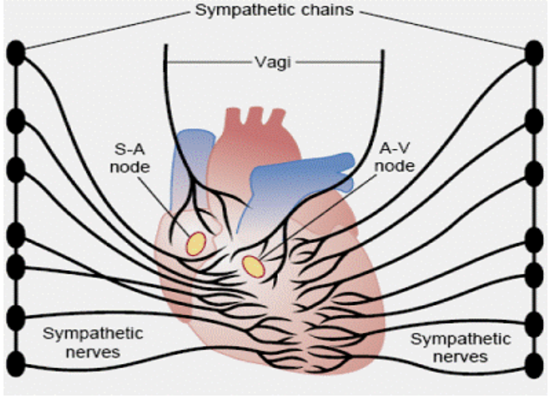


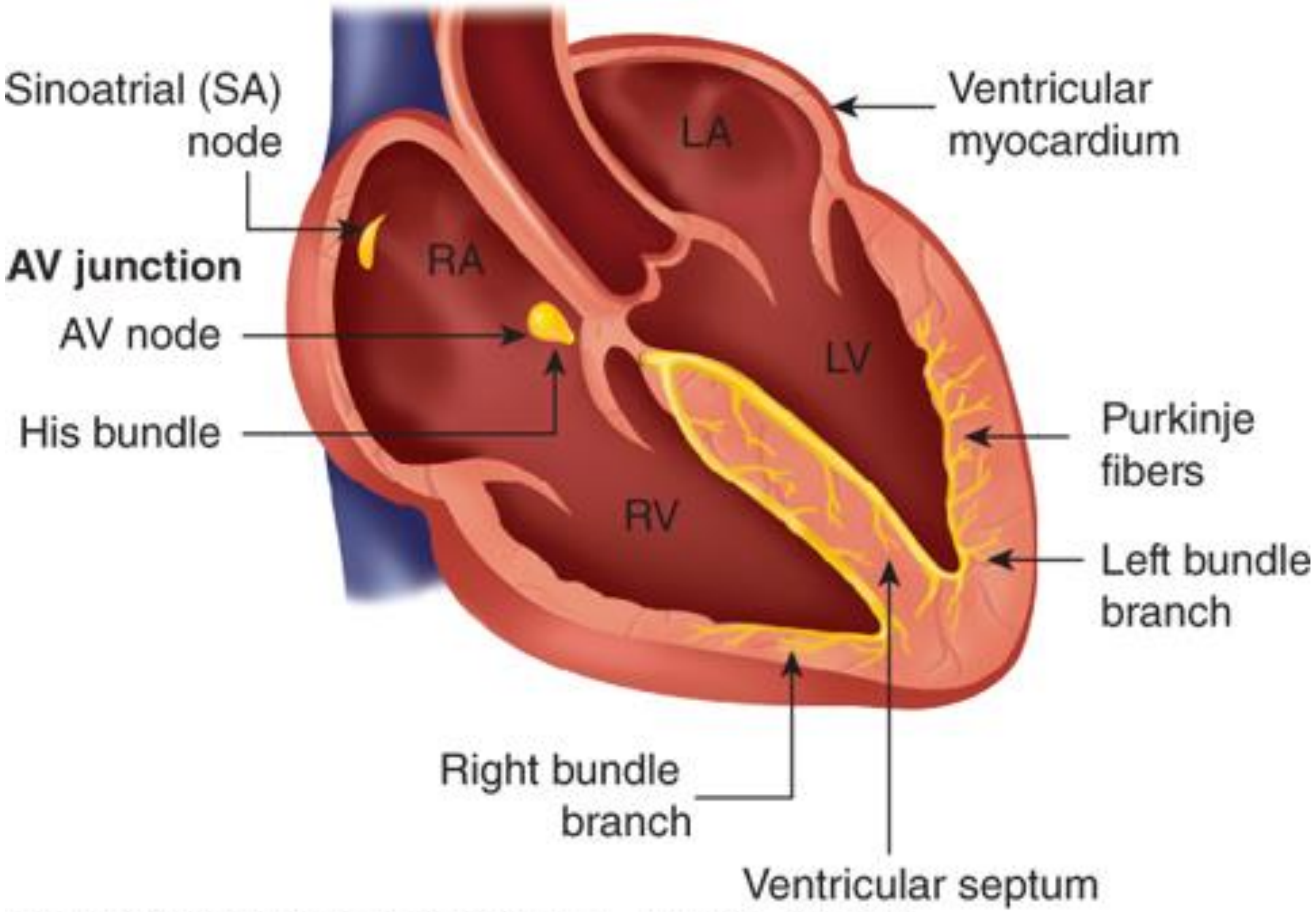


Source: Sylvia C. McKean, John J. Ross, Daniel D. Dressler, Danielle B. Scheurer: Principles and Practice of Hospital Medicine, Second Edition, www.accessmedicine.com Copyright © McGraw-Hill Education. All rights reserved.

P wave: atrial contraction

# Cardiac conduction system

Citation: Chapter 102 Tachycardia, McKean SC, Ross JJ, Dressler DD, Scheurer DB. *Principles and Practice of Hospital Medicine, 2e;* 2017.
Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=146977575&bookid=1872 Accessed: December 29, 2020
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# Potential adverse effects (most are minor)

Bradycardia

Bronchospasm (asthma, COPD) due to β2 block

β2 block may potentiate or mask hypoglycemia in diabetic patients prone to hypoglycemia

- Inhibits reflex hepatic production of glucose
- Masks warning signs (eg, rapid heart beat, tremors, sweating, anxiousness)

Lipid abnormalities: modest effect on triglycerides and HDL-C

Insomnia; fatigue; depression, sexual dysfunction (based on poor studies and subject of debate)

**Abrupt discontinuation of beta blocker after long-term treatment can cause severe angina, myocardial infarction or ventricular arrhythmia, and may increase the risk of sudden death.**

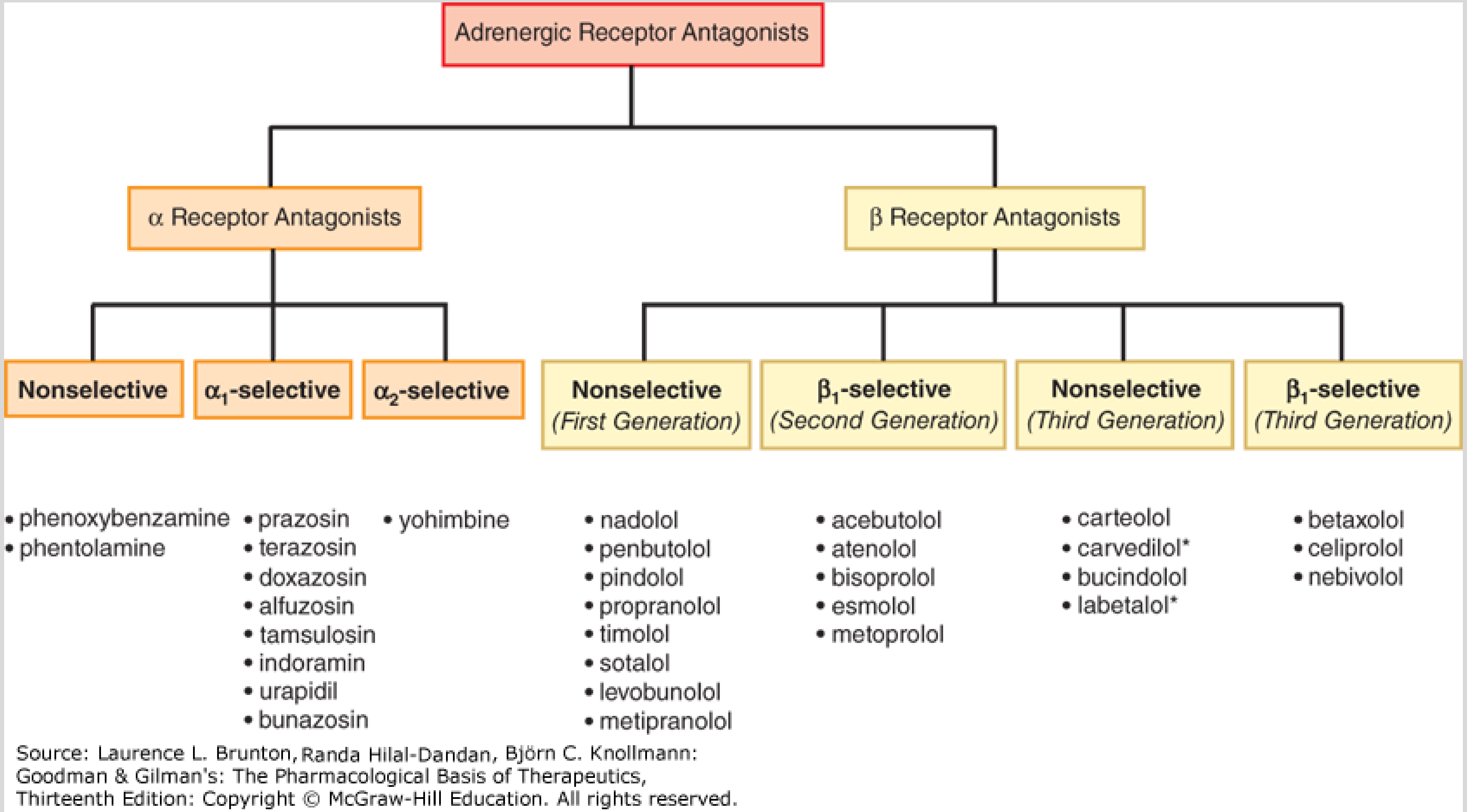
- Supersensitivity mediated by βAR adaptive upregulation
- *Taper the dose of the beta blocker over several weeks.*

# Thought Questions



- How do the beta blockers' inhibitory effects on cardiac cells reduce blood pressure?
- What is the mechanistic thread that connects all the therapeutic uses of the beta blockers?
- How does the mechanism of action describe the mechanisms of adverse effects associated with beta blockers?

# Summary of Adrenergic Antagonists




Adrenergic antagonists inhibit the interaction of endogenous NE and epinephrine, and also sympathomimetic drugs, with adrenergic receptors at effector organs.

Goodman & Gilman's The Pharmacological Basis of Therapeutics, 13e, 2018; Figure 12-3

- The αAR antagonists (alpha blockers) relax vascular smooth muscle and decrease peripheral vascular resistance. They reduce the pressor effects of α agonists and can convert epinephrine's vasopressor effects to a depressor response. Epinephrine reversal is a predictable effect of alpha blockade.
- The βAR antagonists (beta blockers) reduce heart rate, conduction velocity, force of contraction, and cardiac output.
- Both alpha and beta blockers reduce blood pressure.
- The nonselective alpha blockers include phenoxybenzamine (an irreversible inhibitor) and phentolamine (a competitive blocker). Phenoxybenzamine is primarily used to manage presurgical hypertensive and sweating episodes caused by pheochromocytoma. Phentolamine is for rapid, short-term use in intraoperative severe hypertension in pheochromocytoma patients, reversal of extravasation of pressor agents, and reversal of oral soft tissue local anesthesia. They cause marked tachycardia and marked orthostatic hypotension.
- The alpha-1 blockers are used as add-on agents for the management of primary hypertension and they are used for the treatment of urinary symptoms due to benign prostatic hypertrophy (BPH). They cause orthostatic hypotension and syncope (first-dose effect), intraoperative floppy iris syndrome, miosis, nasal stuffiness, and impaired/retrograde ejaculation.

- $\beta_1$-Receptors are primarily found in the heart and kidneys. In the heart, they control heart rate and contractility. Long-term use of a beta blocker reduces blood pressure and myocardial oxygen demand. $\beta_2$-Receptors are primarily found in the lungs, skeletal muscle vasculature, and liver. Blocking $\beta_2$-receptors results in increased airway resistance and metabolic changes that can result in hypoglycemia and hyperkalemia.
- Beta blockers have prominent effects on the heart, decreasing heart rate (chronotropy) and force of contraction (inotropy), and they suppress renin release from specialized cells in the kidney. CNS effects likely contribute to their blood pressure lowering effects.
- Frequently used beta blockers are the nonselective agents that block both $\beta_1$ and $\beta_2$ receptors, cardioselective ($\beta_1$ only) agents, and drugs with combined α and β blocking actions, such as labetalol.
- Beta blockers are useful in the treatment of a number of cardiovascular diseases, such as arrhythmia, effort angina (ischemic heart disease), acute myocardial infarction, congestive heart failure, and as add-on agents in the treatment of hypertension. Beta blockers are also used in the treatment of tachycardia associated with hyperthyroidism, esophageal varices, migraine prophylaxis, and symptoms of performance anxiety.
- Adverse effects include bronchoconstriction ($\beta_2$ blockade), bradycardia, decreased cardiac output, hypoglycemic effects and masking of hypoglycemia, and lipid abnormalities. Depression, fatigue, and sexual dysfunction have been reported.

# Thought Questions



- How does the concept of epinephrine reversal explain the antihypertensive actions of alpha blockers?
- Blocking vascular $\alpha$ARs prevents EPI's vasoconstrictive actions, leading to vasodilation. EPI's $\beta$2-mediated vasodilation is unopposed, contributing to the antihypertensive actions of alpha blockers.
- What is the reason (and mechanism) for advising patients to take the first dose of a selective alpha blocker at bedtime?
- The first dose (or first few doses) may cause orthostatic hypotension because, when rising from a lying down to upright position, reflex vasoconstriction is blocked.
- Selective inhibition of which adrenergic receptor type - $\alpha$1, $\alpha$2, $\beta$1, $\beta$2, or $\beta$3 - leads to tachycardia, hypertension, nausea, vomiting, anorexia, salivation, tremors, diaphoresis, headache, agitation, anxiety, irritability? What drug?

# Thought Questions



- How do the beta blockers' inhibitory effects on cardiac cells reduce blood pressure?
- Beta blockers decrease cardiac output by decreasing heart rate, slowing conduction through the AV node, and decreasing the force of contraction.
- What is the mechanistic thread that connects all the therapeutic uses of the beta blockers?
- Beta blockers inhibit the stimulating effects of the sympathetic nervous system on $\beta$ARs.
- How does the mechanism of action describe the mechanisms of adverse effects associated with beta blockers?
- Bradycardia (all BBs). Blocking the $\beta_2$ARs can exacerbate asthma/COPD and potentiate or mask symptoms of hypoglycemia.